DEPLOYMENT-CONSISTENT PIPELINE FOR WIND POWER FORECASTING AND DISPATCH

Stage 1: DATA LAYER (Corrupted Inputs)

Gaussian noise

Drift bias

Quantization

Structured faults

Operational Context

Stage 2: FORECAST LAYER (Ultra-LSNT Core)

Multi-scale temporal embedding

Moving-average decomposition

Trend branch (lightweight path)

Seasonal branch (nonlinear path, main computation)

Sparse router (Top-K)

Expert pool

Experts

Jump Gate (Conditional skip)

Edge-friendly inference

Stage 3: FEASIBILITY GUIDANCE / INTERFACE LAYER

FG-MOE FEASIBILITY GUIDANCE

Deployable wind proxy

Horizon-aware proxy from **recent** observed wind history

Feasible **power** envelope

Multi-horizon forecast trajectory

Feasibility penalty during training

FIXED SHARED FORECAST-TO-DISPATCH INTERFACE

Affine mapping

Deterministic clipping

No model-specific retuning

Ensures fairness, standardization, same downstream algebra

Stage 4: DISPATCH / VALUE LAYER (Operational Impact)

DAY-AHEAD DISPATCH SCREENING

RTS-24 DC-OPF check

FALLBACK / FEASIBILITY RESTORATION

OPERATING COST

WIND CURTAILMENT

FALLBACK BURDEN